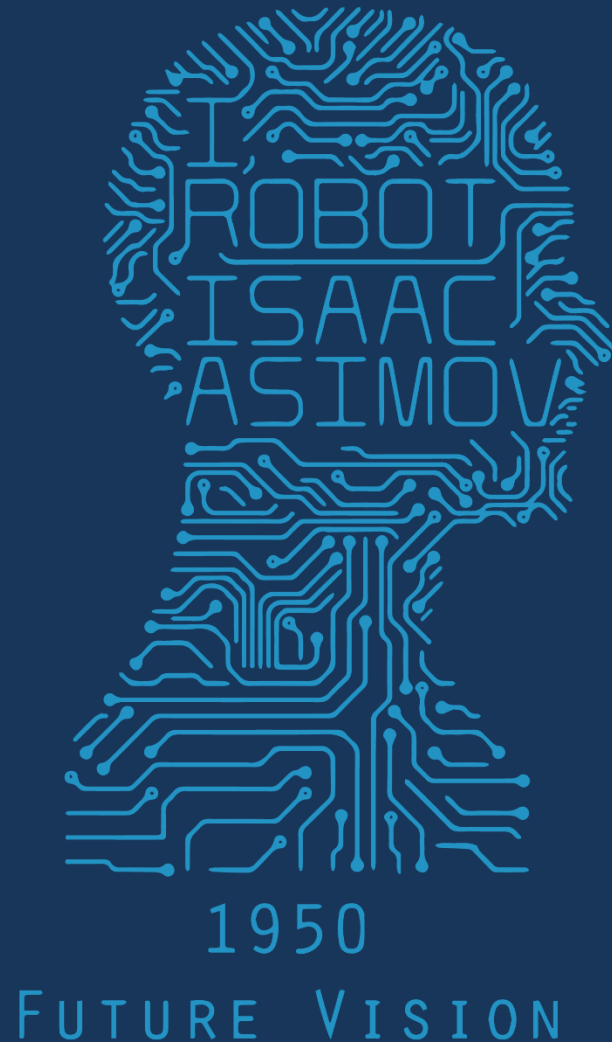


Computer Vision and Pattern Recognition



计算机视觉与 模式识别

Light and Color

Today's Class

- **What is Light**
(with a low-light imaging case for AUV)
- What is Color?
(with an automatic colorization case)

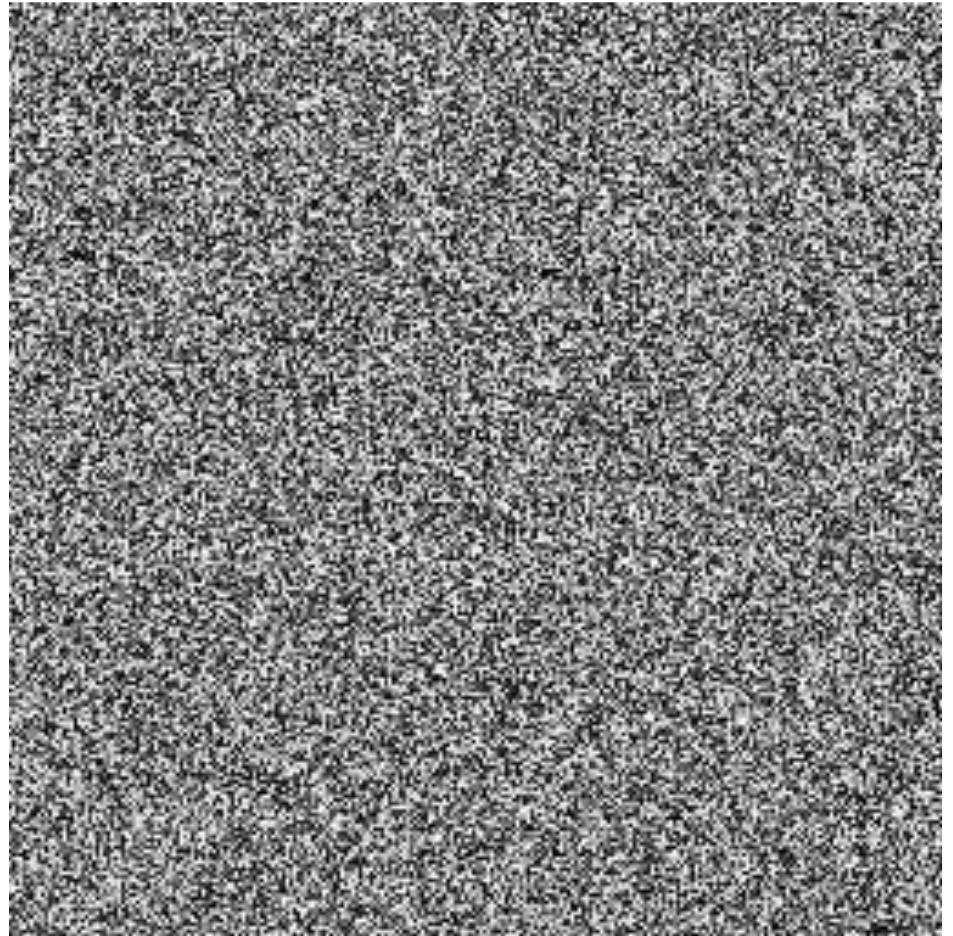
WHAT IS AN IMAGE?

```
>> I = rand(256,256);
```

Think-Pair-Share:

- What is this? What does it look like?
- Which values does it take?
- How many values can it take?
- Is it an image?

```
>> I = rand(256,256);  
>> imshow(I);
```



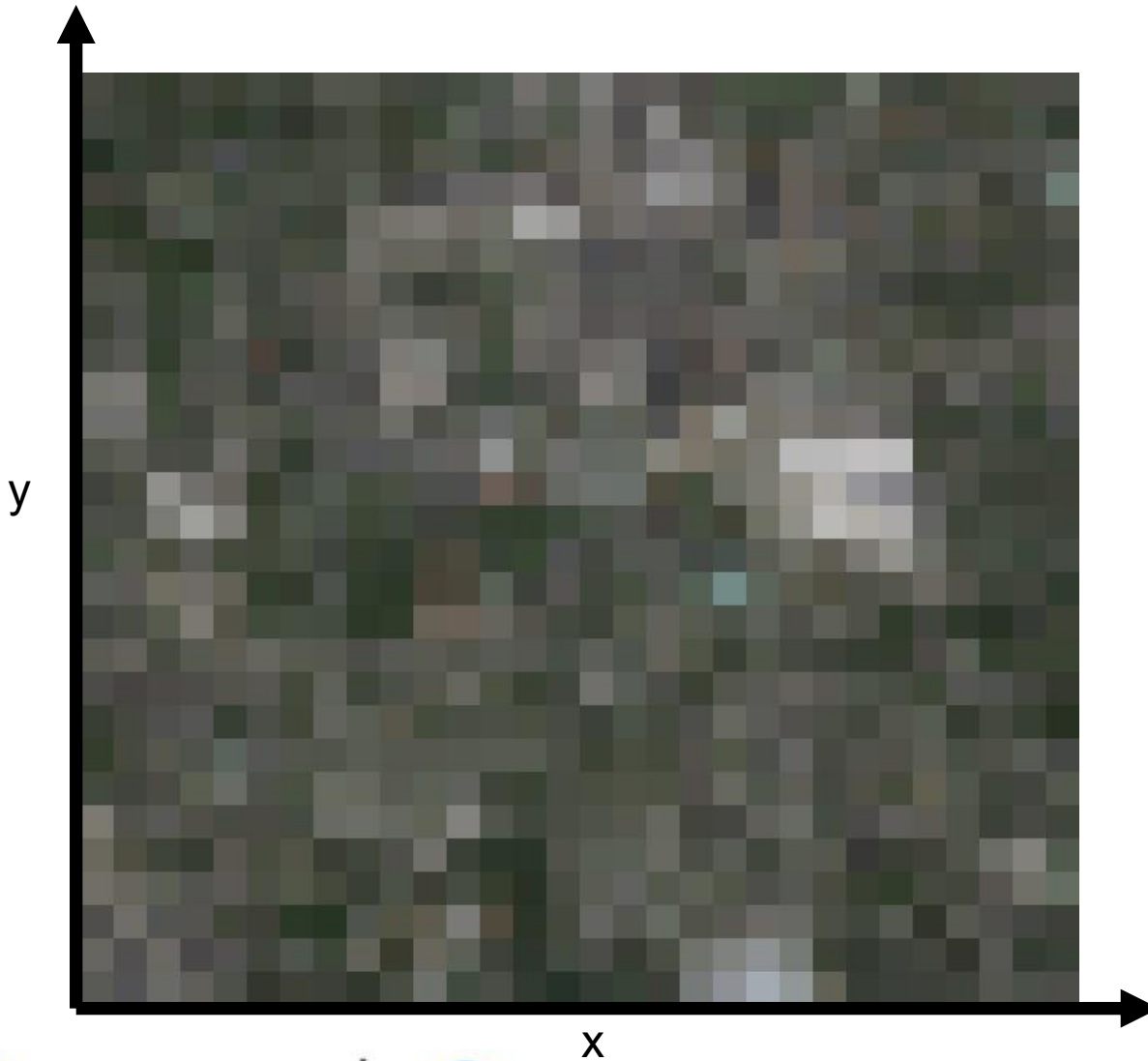
Dimensionality of an Image

- @ 8bit = 256 values ^ 65,536
 - Computer says 'Inf' combinations.
- Some depiction of all possible scenes would fit into this memory.

Dimensionality of an Image

- @ 8bit = $256 \text{ values}^{\wedge} 65,536$
 - Computer says ‘Inf’ combinations.
- Some depiction of all possible scenes would fit into this memory.
- Computer vision as making sense of an extremely high-dimensional space.
 - Subspace of ‘natural’ images.
 - Deriving low-dimensional, explainable models.

What is each part of an image?



What is each part of an image?

- Pixel -> picture element

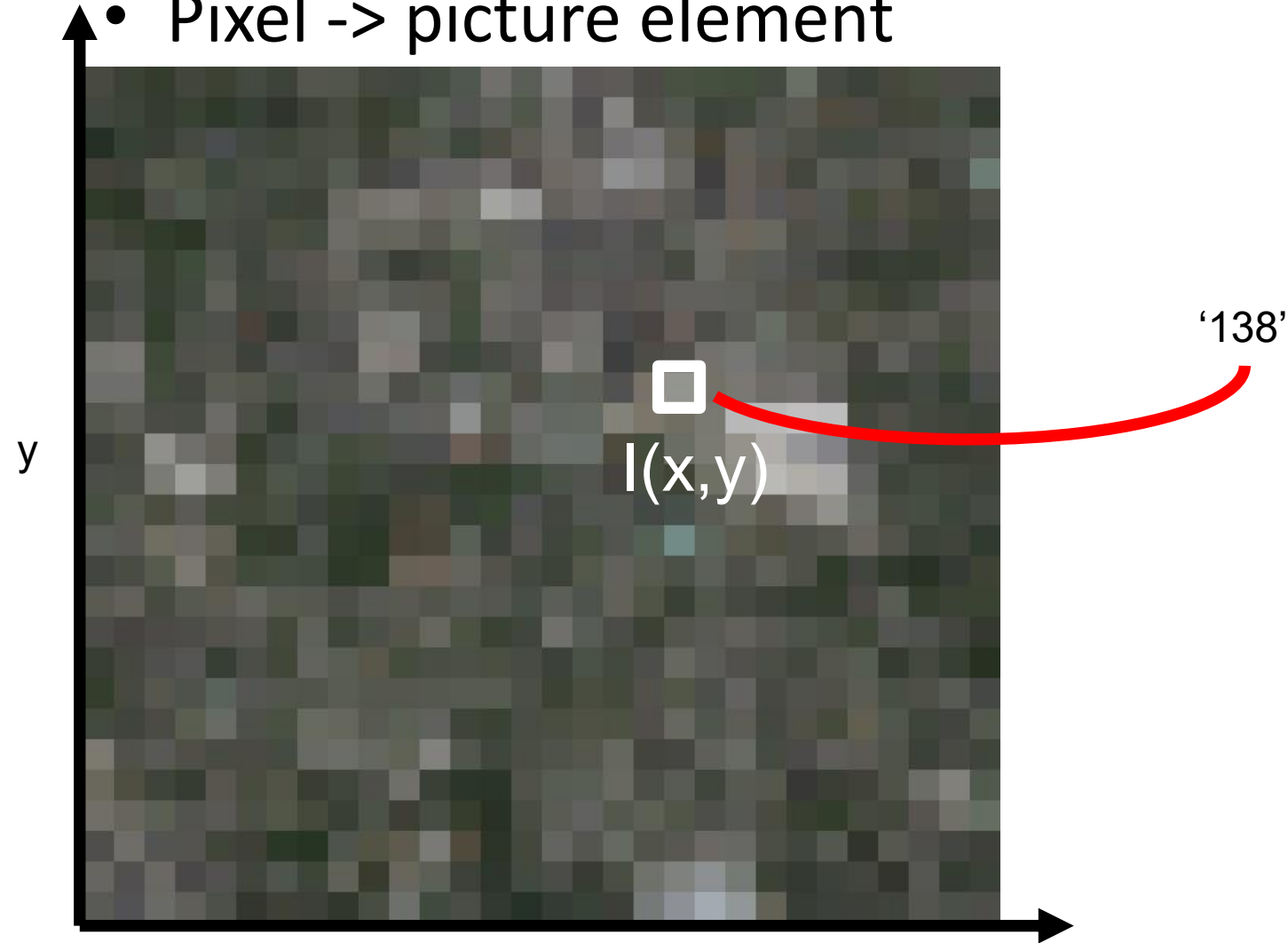
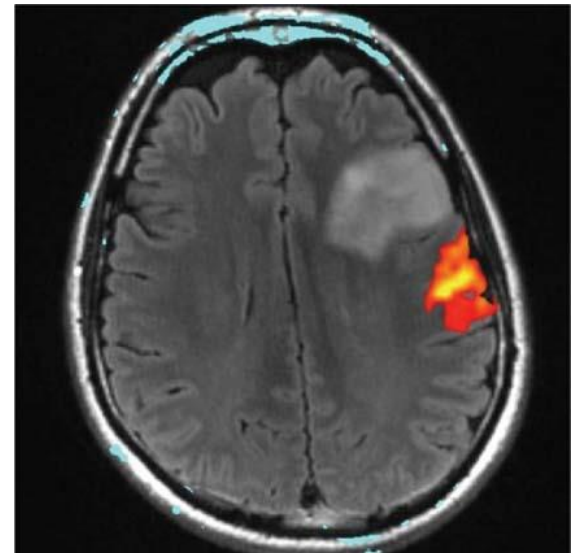
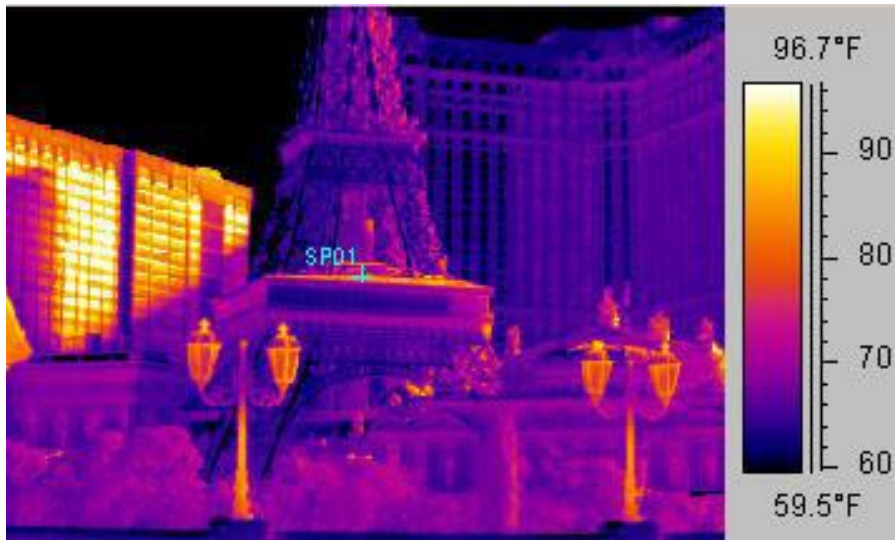


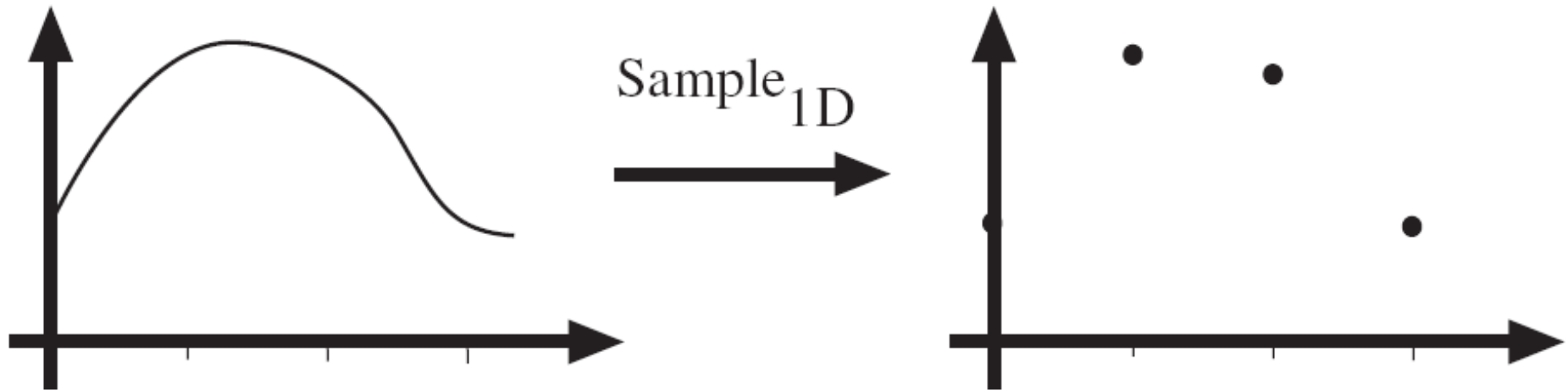
Image as a 2D sampling of signal

- Signal: function depending on some variable with physical meaning.
- Image: sampling of that function.
 - 2 variables: xy coordinates
 - 3 variables: xy + time (video)
 - 'Brightness' is the value of the function for visible light
- Can be other physical values too: temperature, pressure, depth ...

Example 2D Images

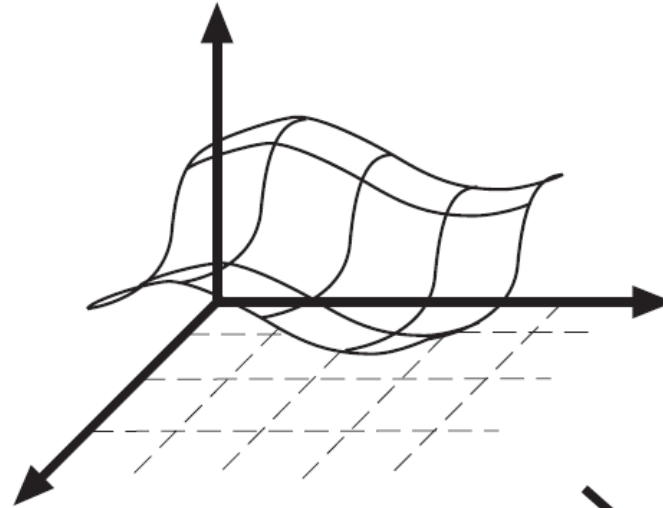


Sampling in 1D

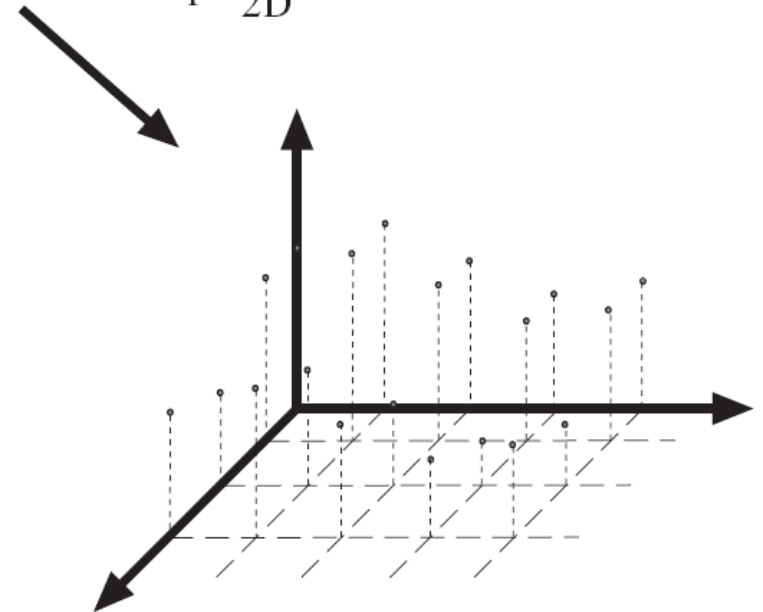


- Sampling in 1D takes a function, and returns a vector whose elements are values of that function at the sample points.

Sampling in 2D

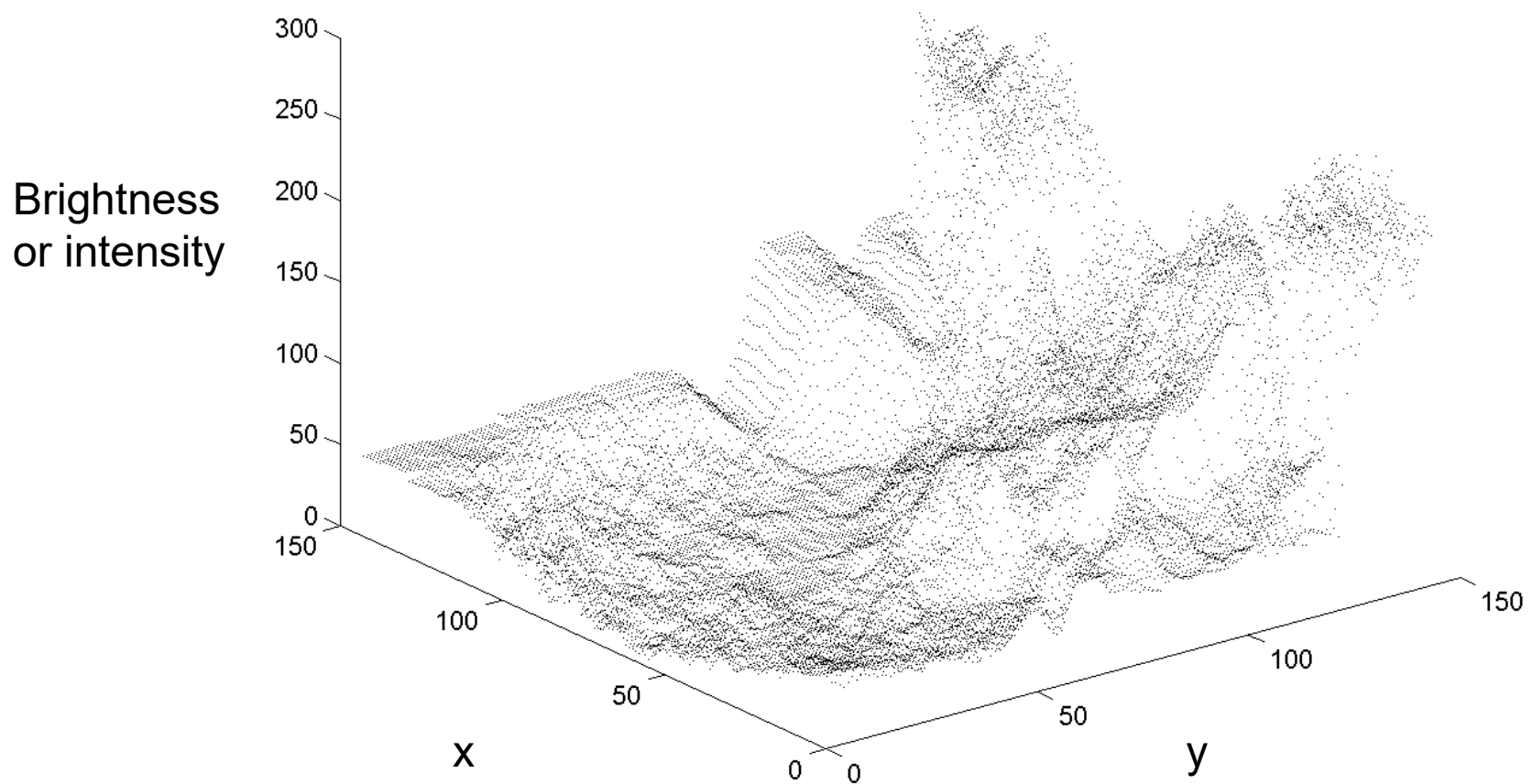


Sample_{2D}



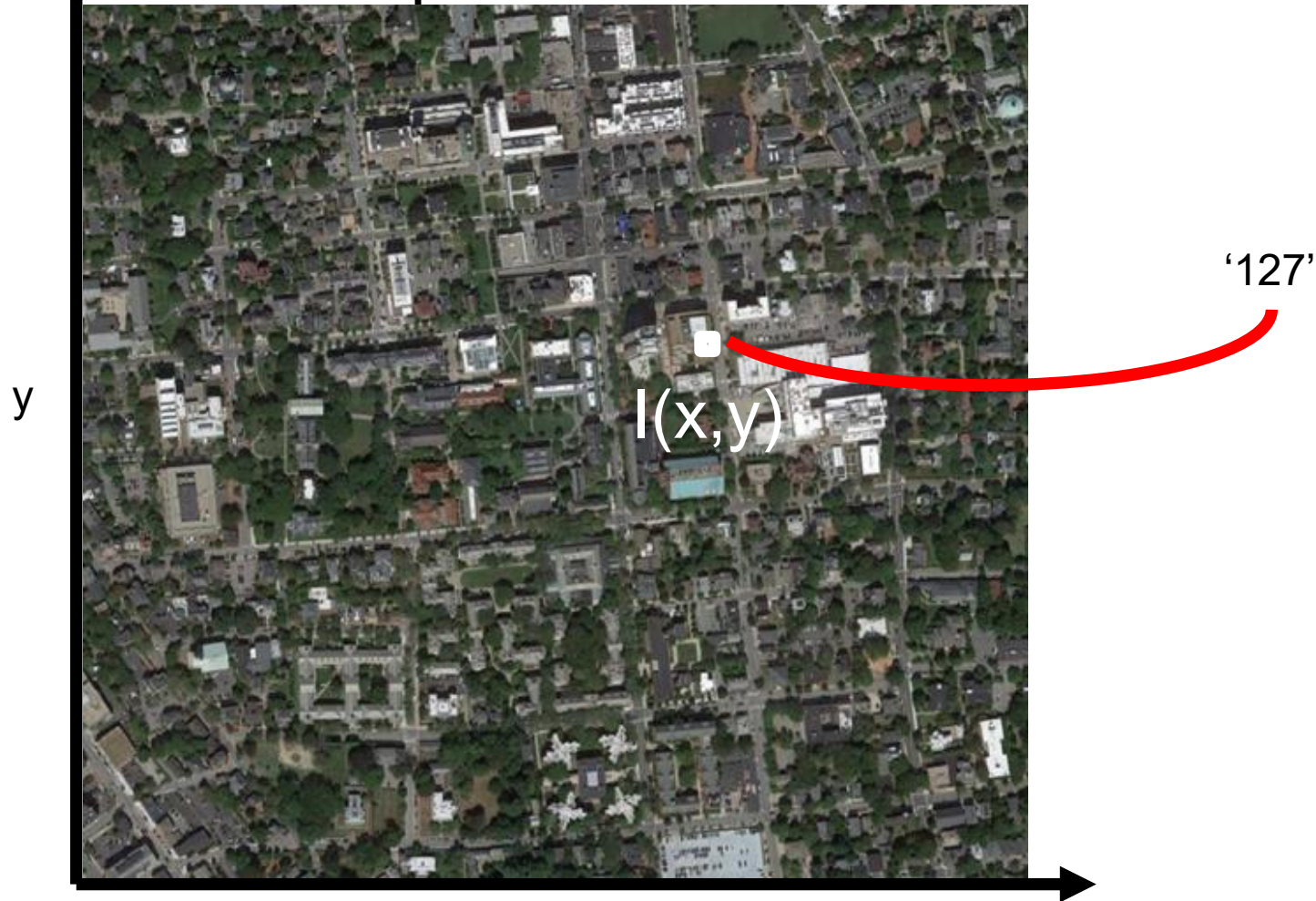
- Sampling in 2D takes a function and returns a matrix.

Grayscale Digital Image

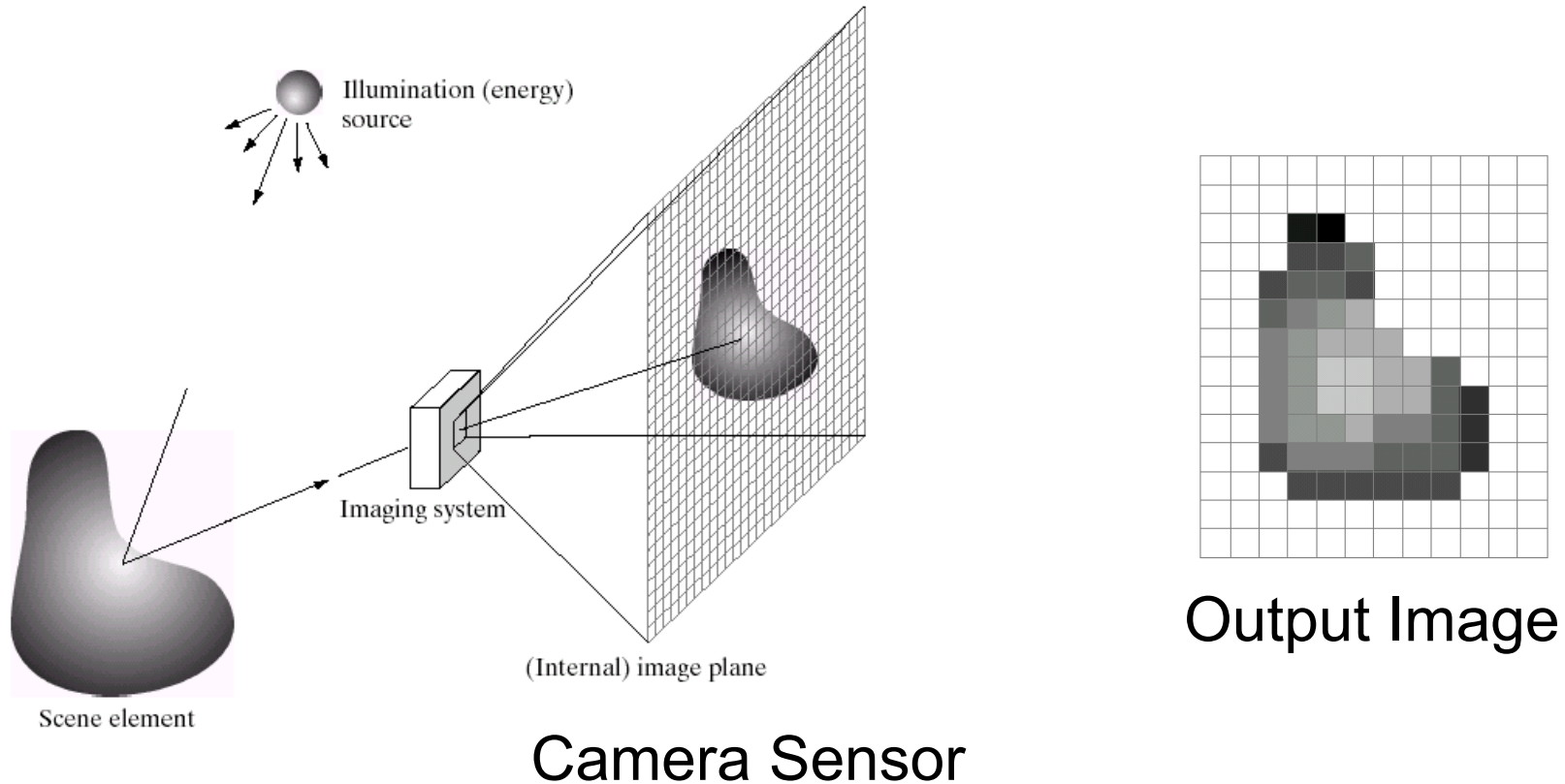


What is each part of a photograph?

- Pixel -> picture element



Integrating light over a range of angles.



But what is color?

ANATOMY

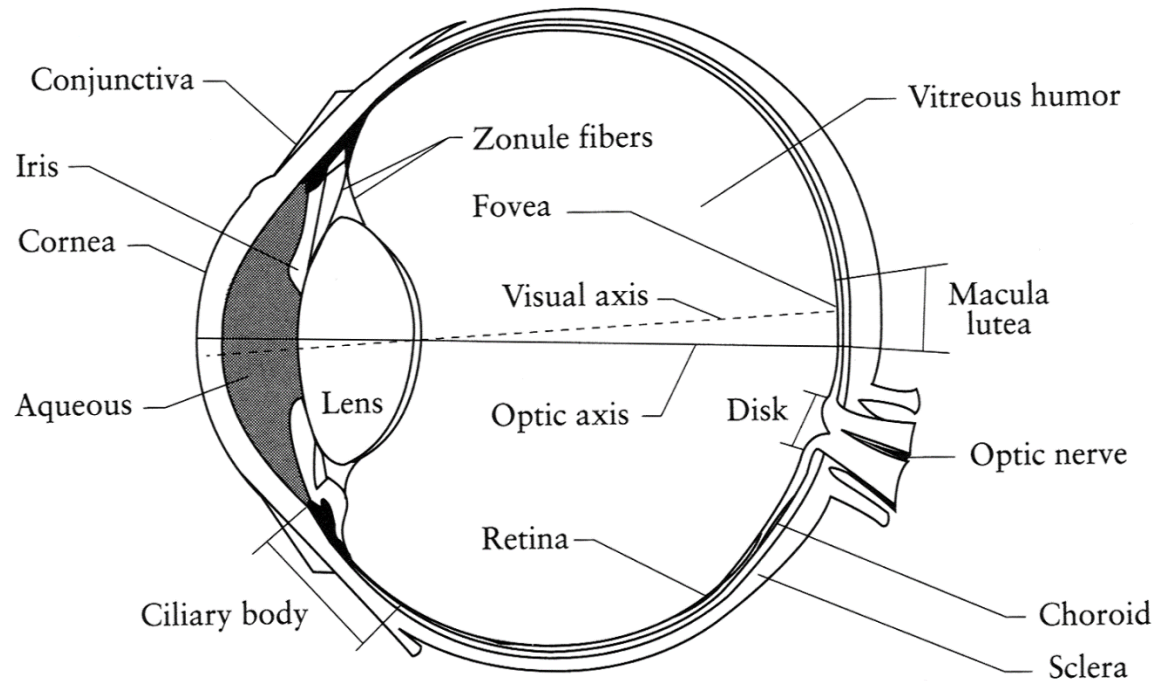


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University of Electronic Science and Technology of China

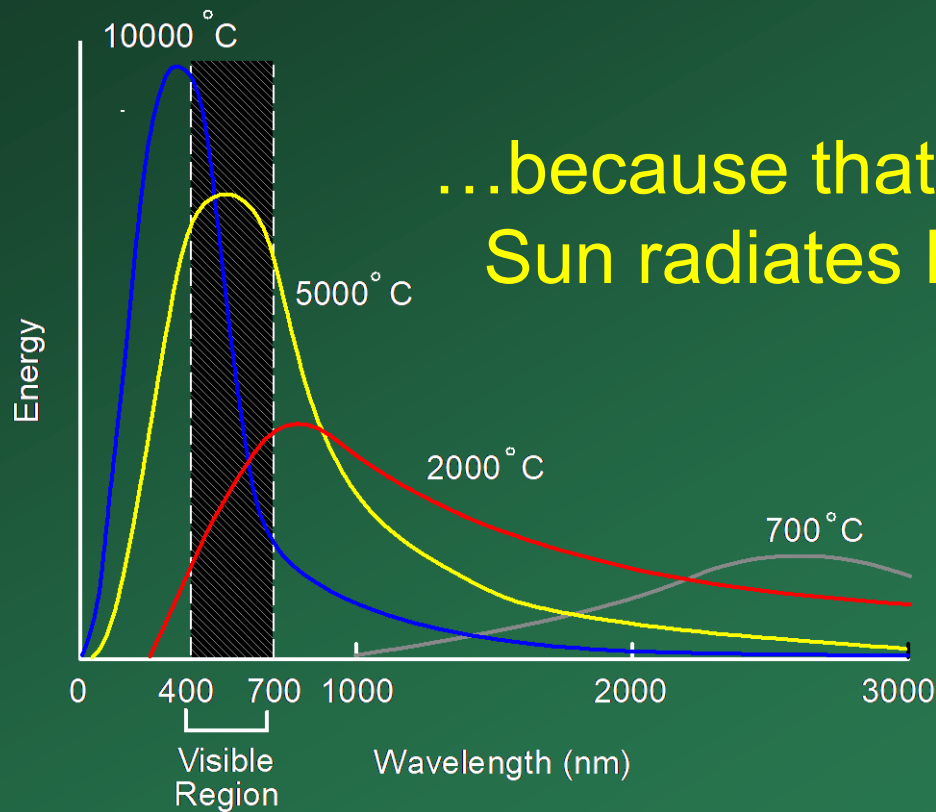
The Eye



- The human eye is a camera
 - **Iris** (虹膜) - colored annulus with radial muscles
 - **Pupil** (瞳孔) - the hole (aperture) whose size is controlled by the iris
 - **What's the sensor?**
 - photoreceptor cells (rods and cones) in the **retina** (视网膜)

Visible Light

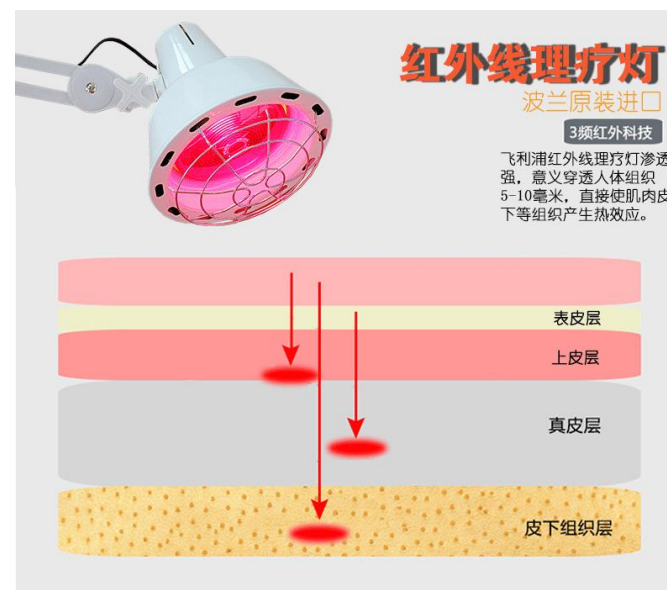
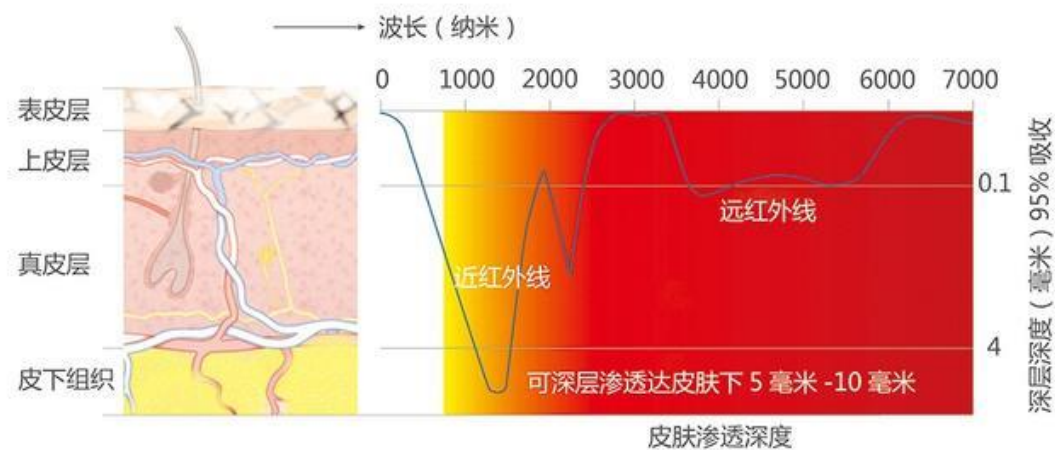
Why do we see light of these wavelengths?



© Stephen E. Palmer, 2002



Visible Light

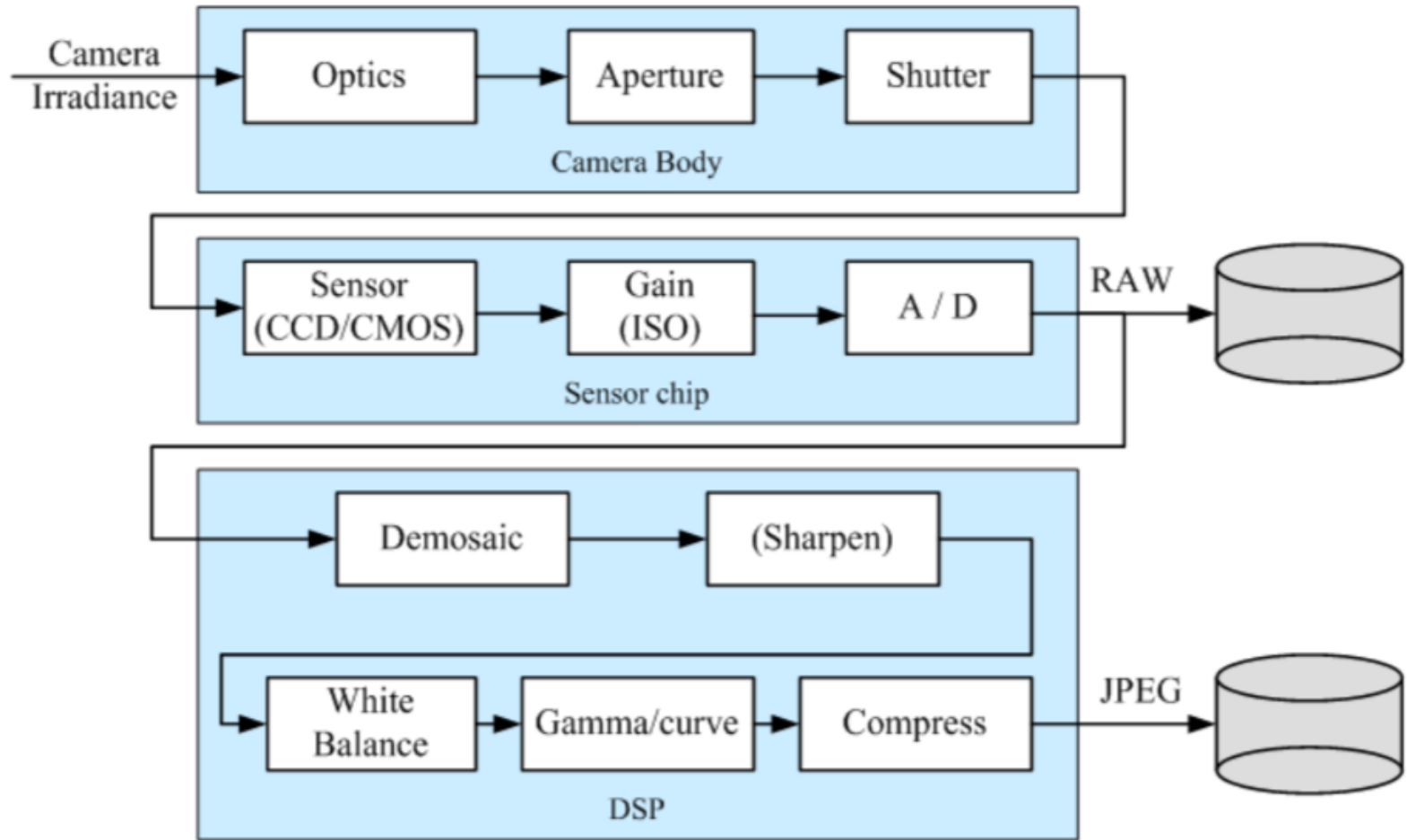


The Digital camera

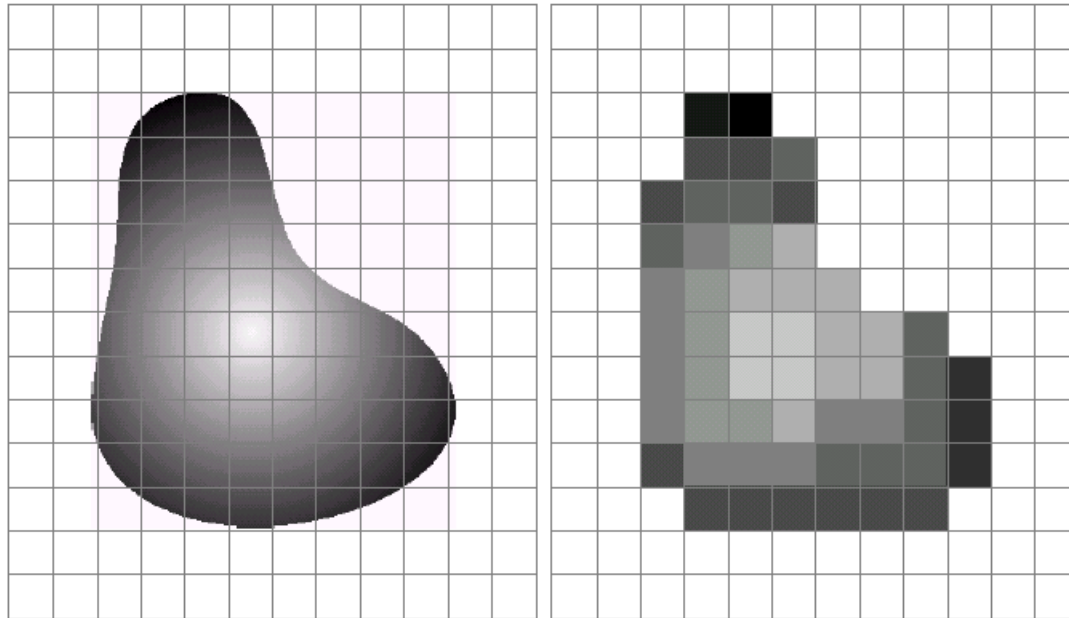


- A digital camera replaces film with a sensor array
 - Each cell in the array is light-sensitive diode that converts photons to electrons
 - Two common types:
 - Charge Coupled Device (CCD)
 - CMOS

The sensing pipeline



Sensor Array



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

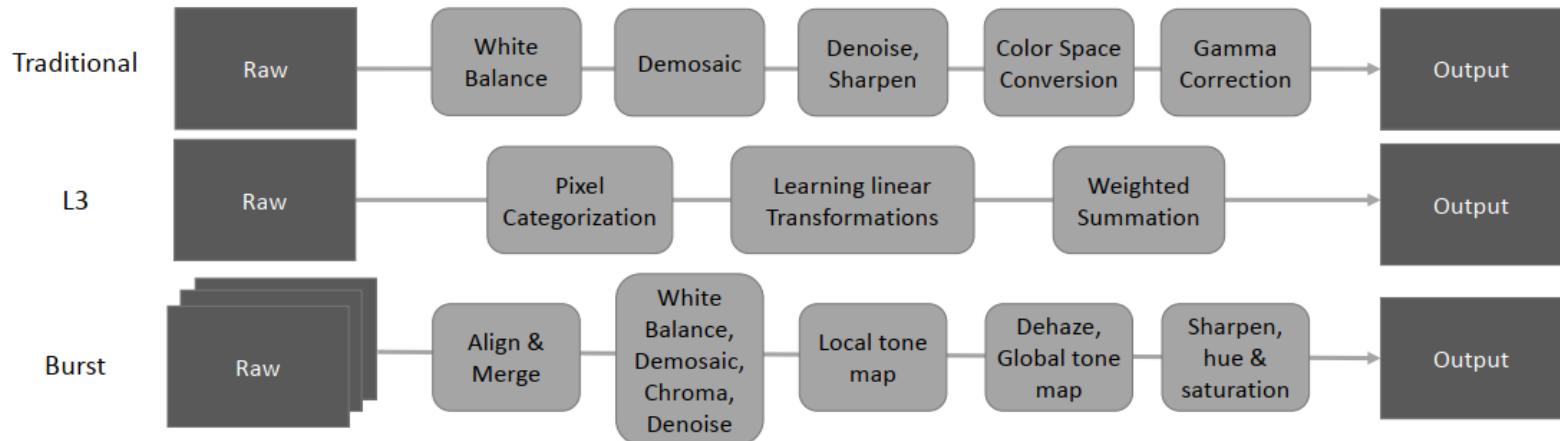


CMOS sensor

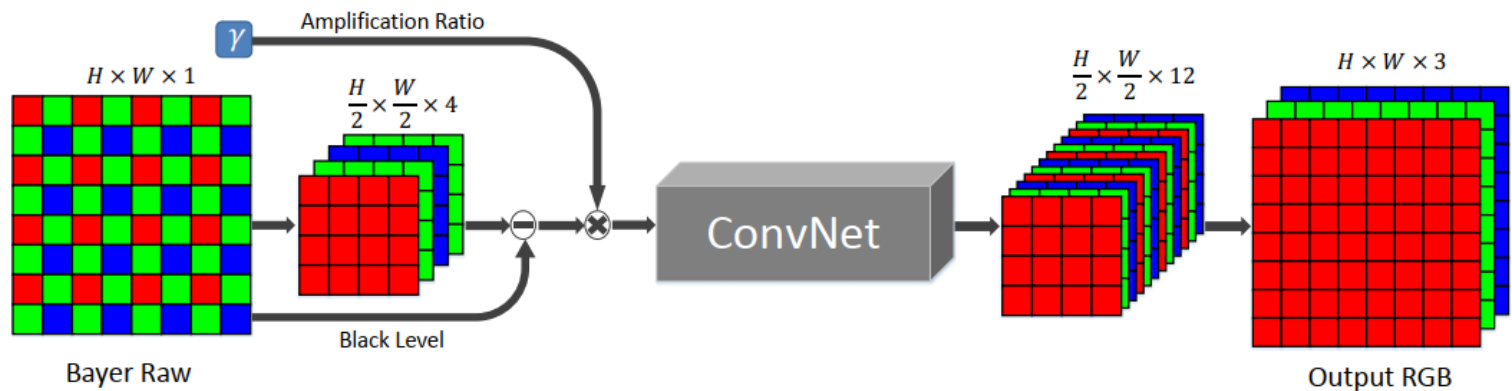
What happens when in Night



Extension-Learning to See in the Dark



(a)



(b)

Chen, C., Chen, Q., Xu, J., & Koltun, V. (2018). Learning to see in the dark. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 3291-3300).

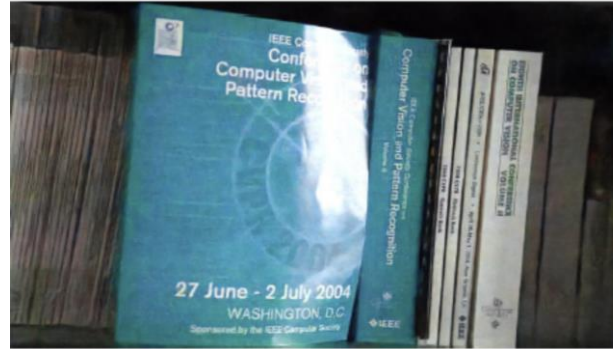
Extension-Learning to See in the Dark



放大率决定了输出的亮度。在我们的管道中，**放大率是外部设置的**，并作为管道的输入提供，**类似于摄像机中的ISO设置**。上图显示了不同放大率的影响。用户可以通过设置不同的放大系数来调整输出图像的亮度。

Chen, C., Chen, Q., Xu, J., & Koltun, V. (2018). Learning to see in the dark. In *Proceedings of the **IEEE conference on computer vision and pattern recognition*** (pp. 3291-3300).

Extension-Learning to See in the Dark



<https://cchen156.github.io/SID.html>

Today's Class

- What is Light
(with a low-light imaging case for AUV)
- **What is Color?**
(with an automatic colorization case)

Color

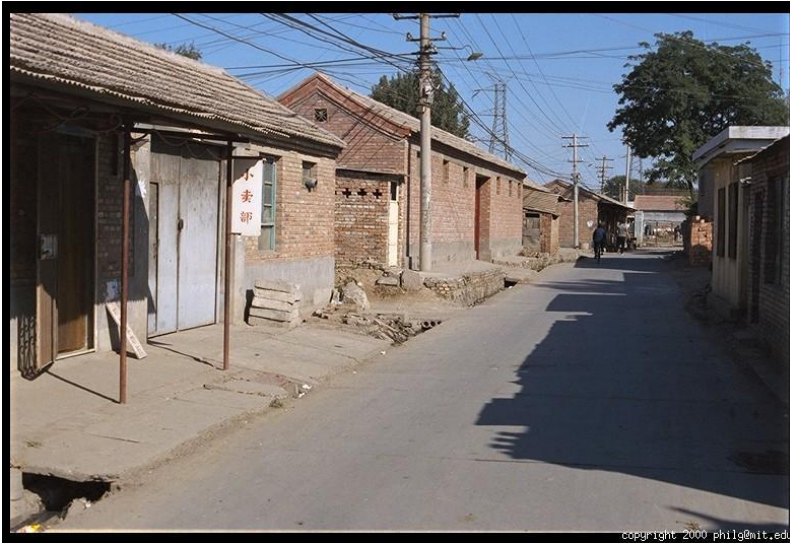
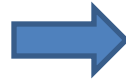
R



G

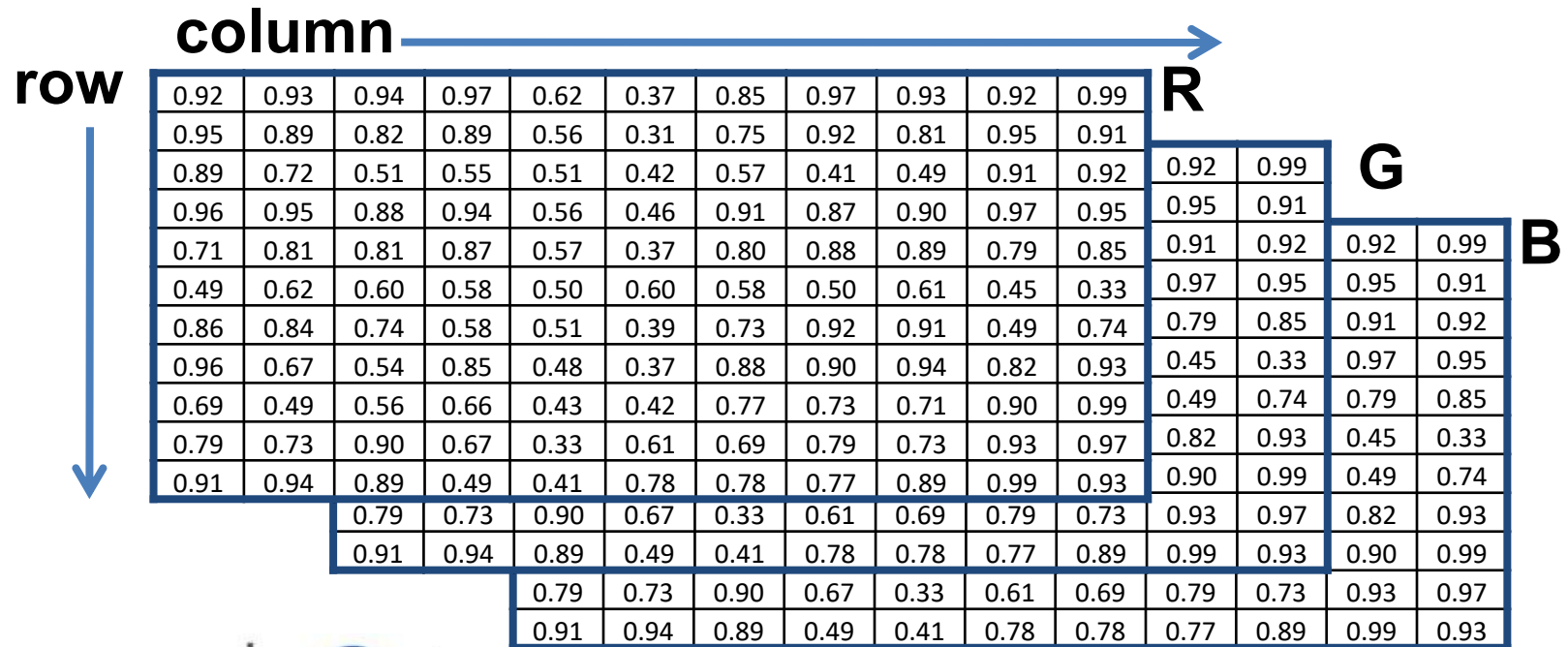


B



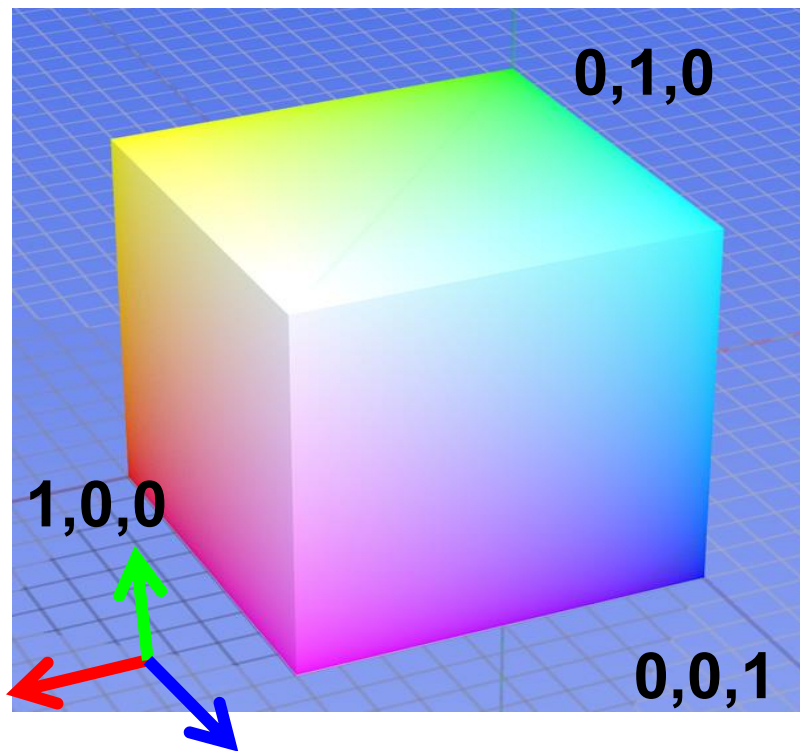
Images in Matlab

- NxM RGB “im”
 - `im(1,1,1)` = top-left pixel value in R-channel
 - `im(y, x, b)` = y pixels down, x pixels to right in the bth channel
 - `im(N, M, 3)` = bottom-right pixel in B-channel
- `imread(filename)` returns a uint8 image (values 0 to 255)
 - Convert to double format (values 0 to 1) with `im2double`



Color spaces: RGB

Default color space



Any color = $r \cdot R + g \cdot G + b \cdot B$

- Strongly correlated channels
- Non-perceptual



R = 1
(G=0,B=0)



G = 1
(R=0,B=0)



B = 1
(R=0,G=0)

Got it. $C = r \cdot R + g \cdot G + b \cdot B$

IS COLOR A VECTOR SPACE?
THINK-PAIR-SHARE



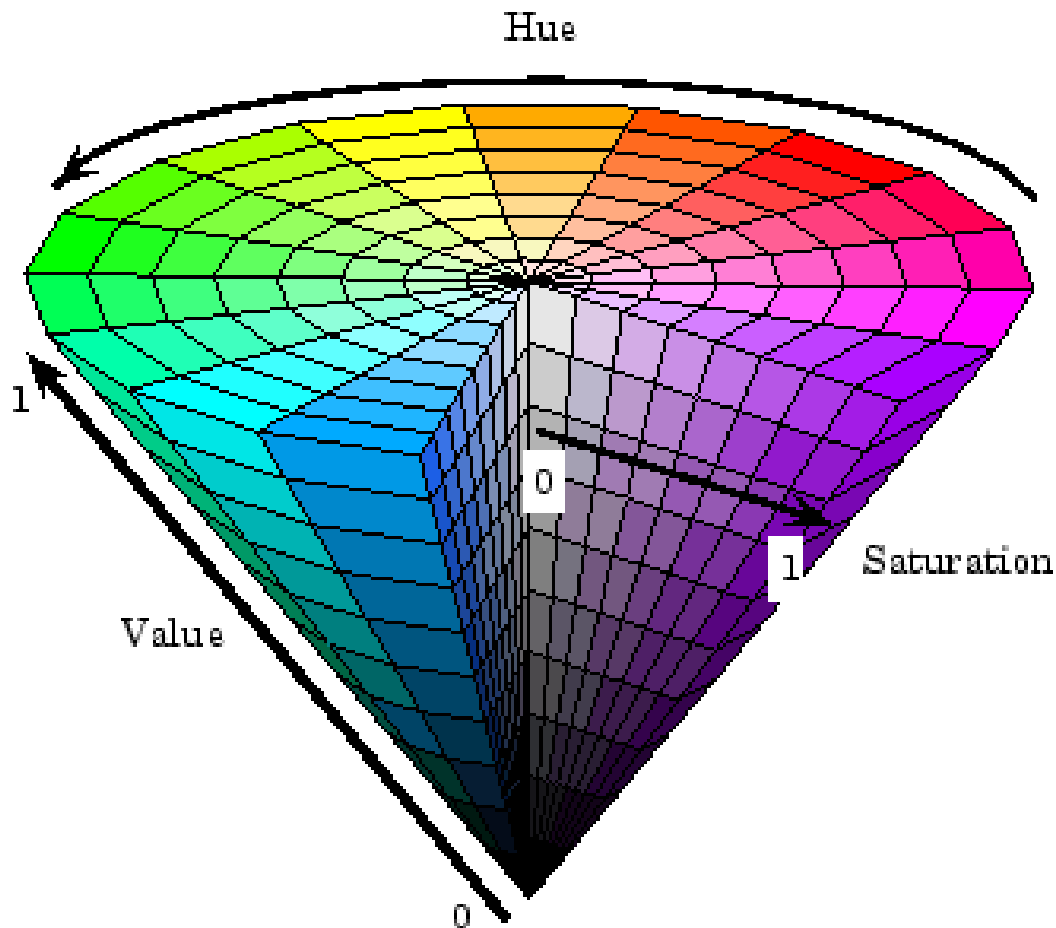
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Color spaces: HSV

Intuitive color space



If you had to choose, would you rather go without:

- intensity ('value'), or
- hue + saturation ('chroma')?

Think-Pair-Share

Most information in intensity



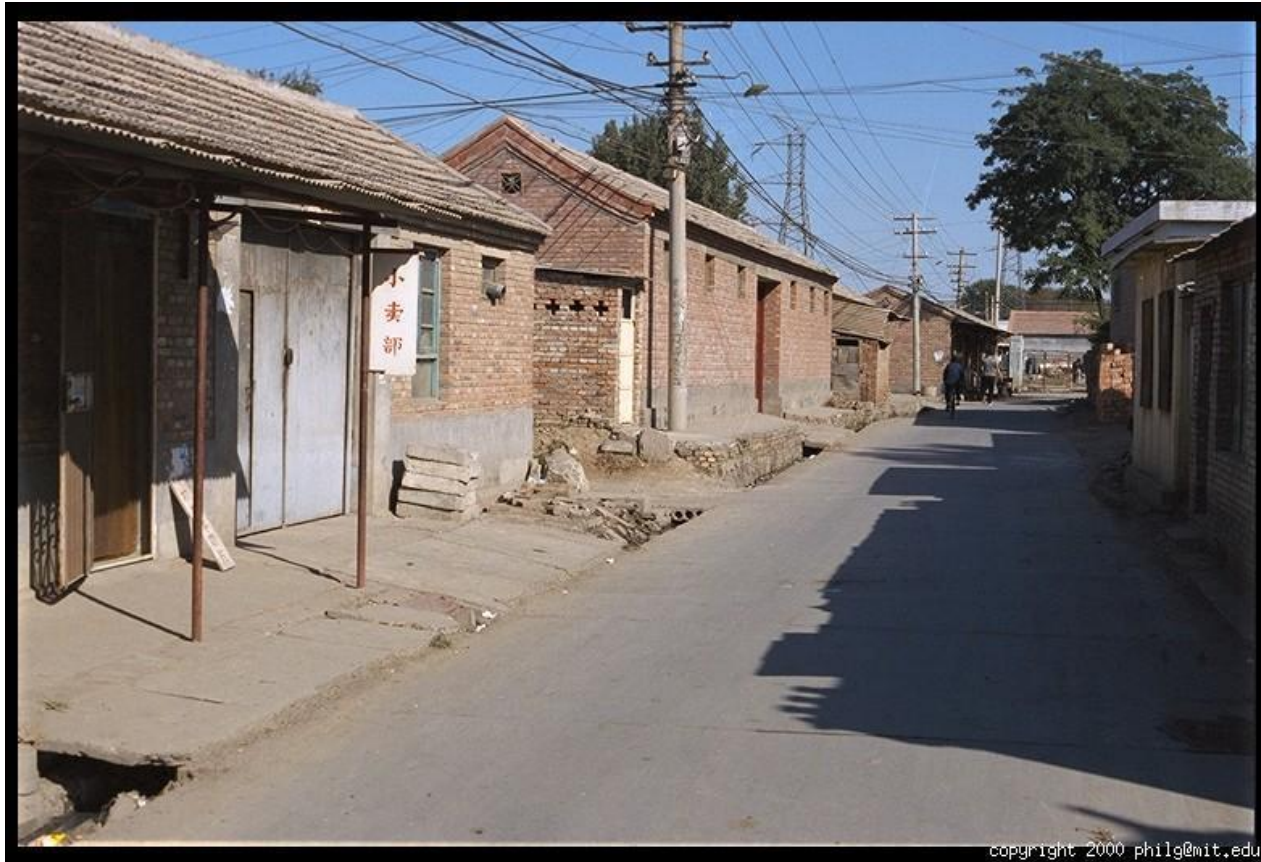
Only color shown – constant intensity

Most information in intensity



Only intensity shown – constant color

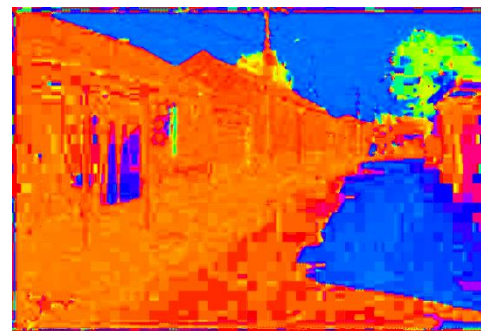
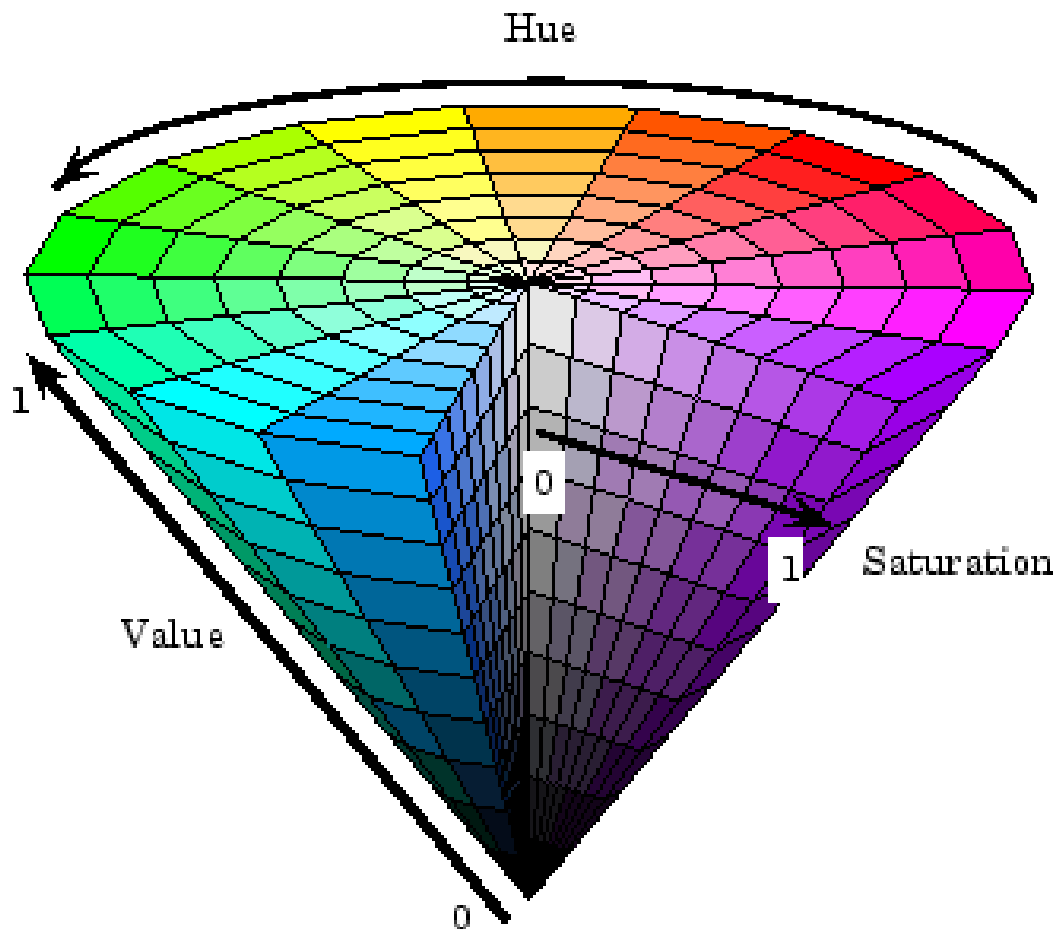
Most information in intensity



Original image

Color spaces: HSV

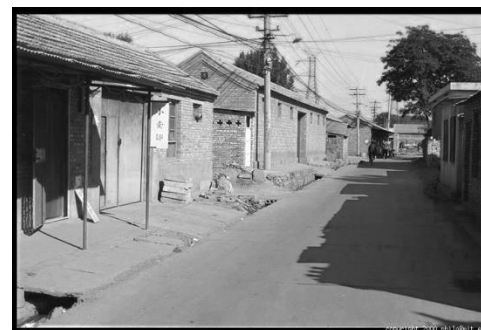
Intuitive color space



H
(S=1,V=1)



S
(H=1,V=1)



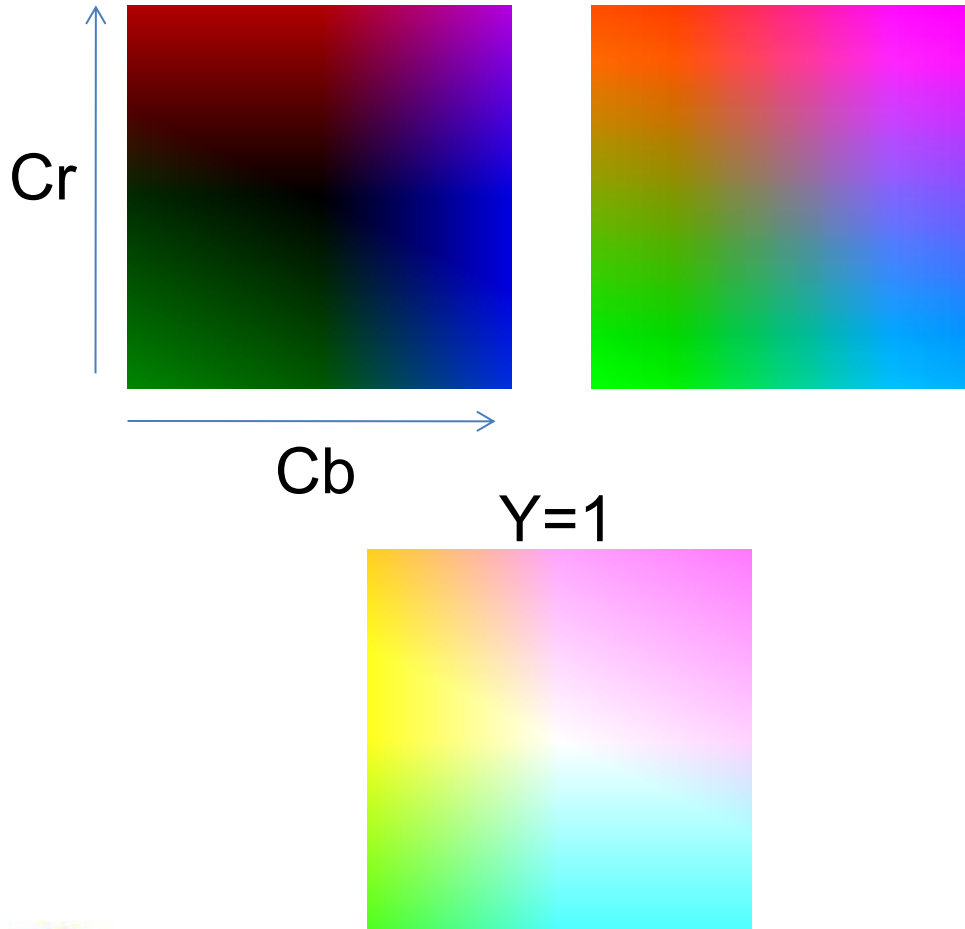
V
(H=1,S=0)

Color spaces: YCbCr

Fast to compute, good for compression, used by TV

Y=0

Y=0.5



Y
(Cb=0.5,Cr=0.5)



Cb
(Y=0.5,Cr=0.5)



Cr
(Y=0.5,Cb=0.5)

More references

- <https://www.colorsystem.com/>
- A description of many different color systems developed through history.
- Navigate from the right-hand links.
- Thanks to Alex Nibley!